OpenClaw Temperature and Humidity Reading Tutorial

> Before starting, please complete the following in the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05` for now. This article assumes the basic environment is already prepared and only covers DHT-related content.

1. Tutorial Objective

This article demonstrates OpenClaw's natural language control capabilities through the DHT temperature and humidity sensor. After completing this tutorial, readers will be able to:

- Understand the basic approach of OpenClaw reading temperature and humidity
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice interaction
- Complete four DHT cases: single reading, comfort judgment, threshold alert, and OLED linked display
- Understand the complete chain from "user question" to "sensor reading" and "result feedback"

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor `[Default DHT11]`
- `[Optional] OLED display`
- `[Optional] Microphone module`
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Low-level driver code: `/home/jetson/hardware_hal_demo/`
- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- Temperature and humidity linked display: `/home/jetson/hardware_hal_demo/dht_oled`

- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

The `dht` tool supports the following capabilities:

- Read current temperature and humidity
- Output results in `json` or `text` format
- Use `mock` fake data for local debugging

```
dht_oled  #Display temperature and humidity data on OLED
dht #Return `json` result
```

If you want to display temperature and humidity results directly on the OLED, you can also use `dht_oled`.

Note: If you want to write your own script to read temperature and humidity data using the ajetson Python driver library, it is recommended to read DHT data at 2-second intervals, as the DHT sensor works best with a 2-second sampling period.

4. Testing Temperature and Humidity Reading

Before starting this tutorial, please complete a basic `openclaw tui` dialogue self-check first; if you haven't done so, see [06-Three Dialogue Configuration Tests]. After confirming that OpenClaw can reply normally, proceed to the DHT-specific test.

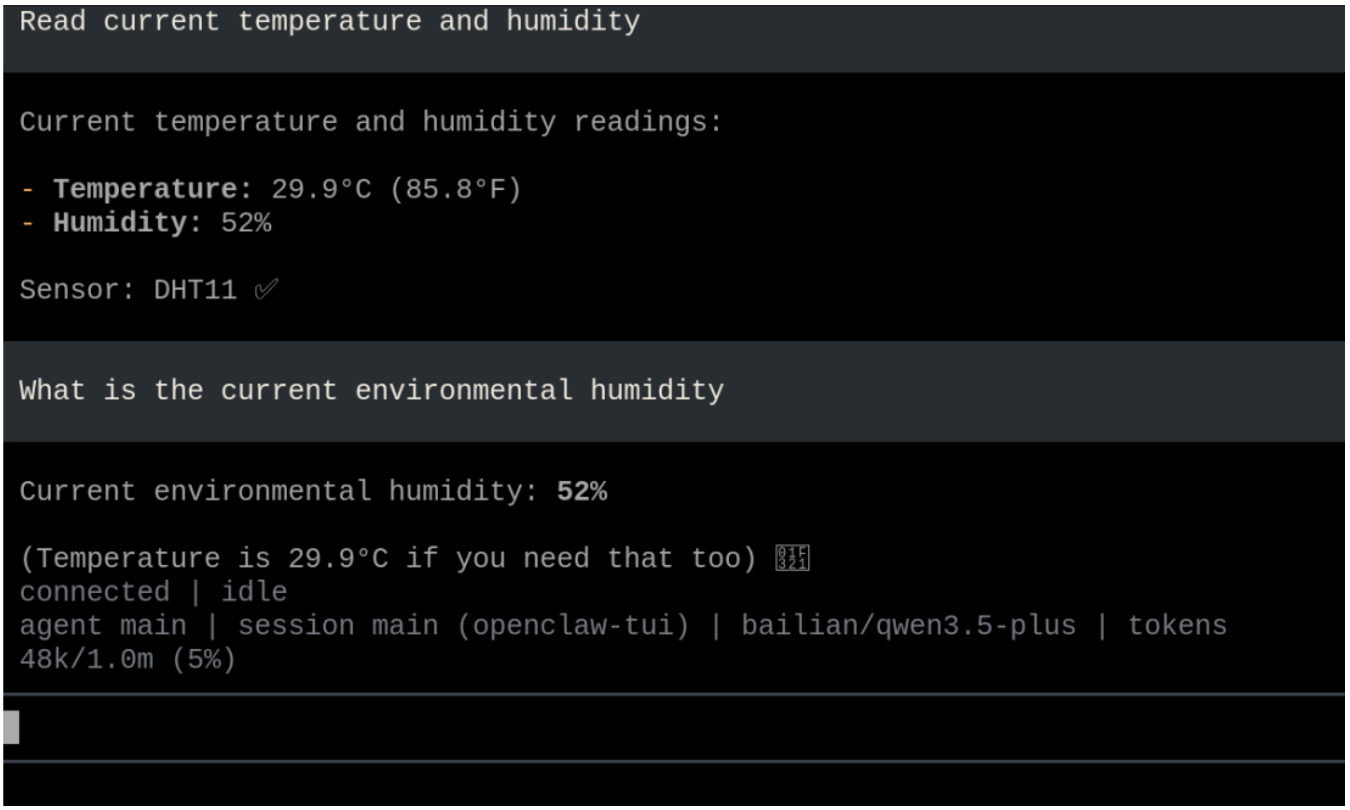
Enter the following command in the terminal to enter TUI:

```
openclaw tui
```



You can try entering the following content in the terminal:

- Read the current temperature and humidity
- What is the current ambient humidity
- Help me judge whether the current environment is comfortable
- Alert me if the temperature is too high



5. Three Schemes for Communicating with OpenClaw

Scheme	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote control, classroom demonstration	Easy to connect, suitable for multi-person experience	Demonstrate "remote natural language control"
TUI	Local debugging, command verification	Most direct, easiest to troubleshoot	First run through the control chain
Voice code	Voice interaction, complete AI experience	Closest to a real conversational experience	As the final demonstration scheme

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

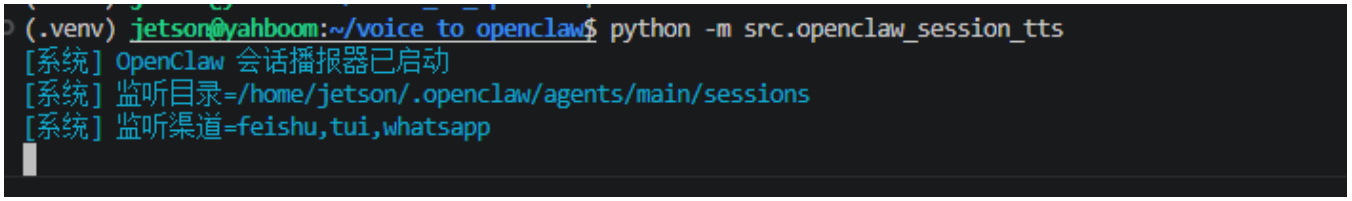
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After
modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

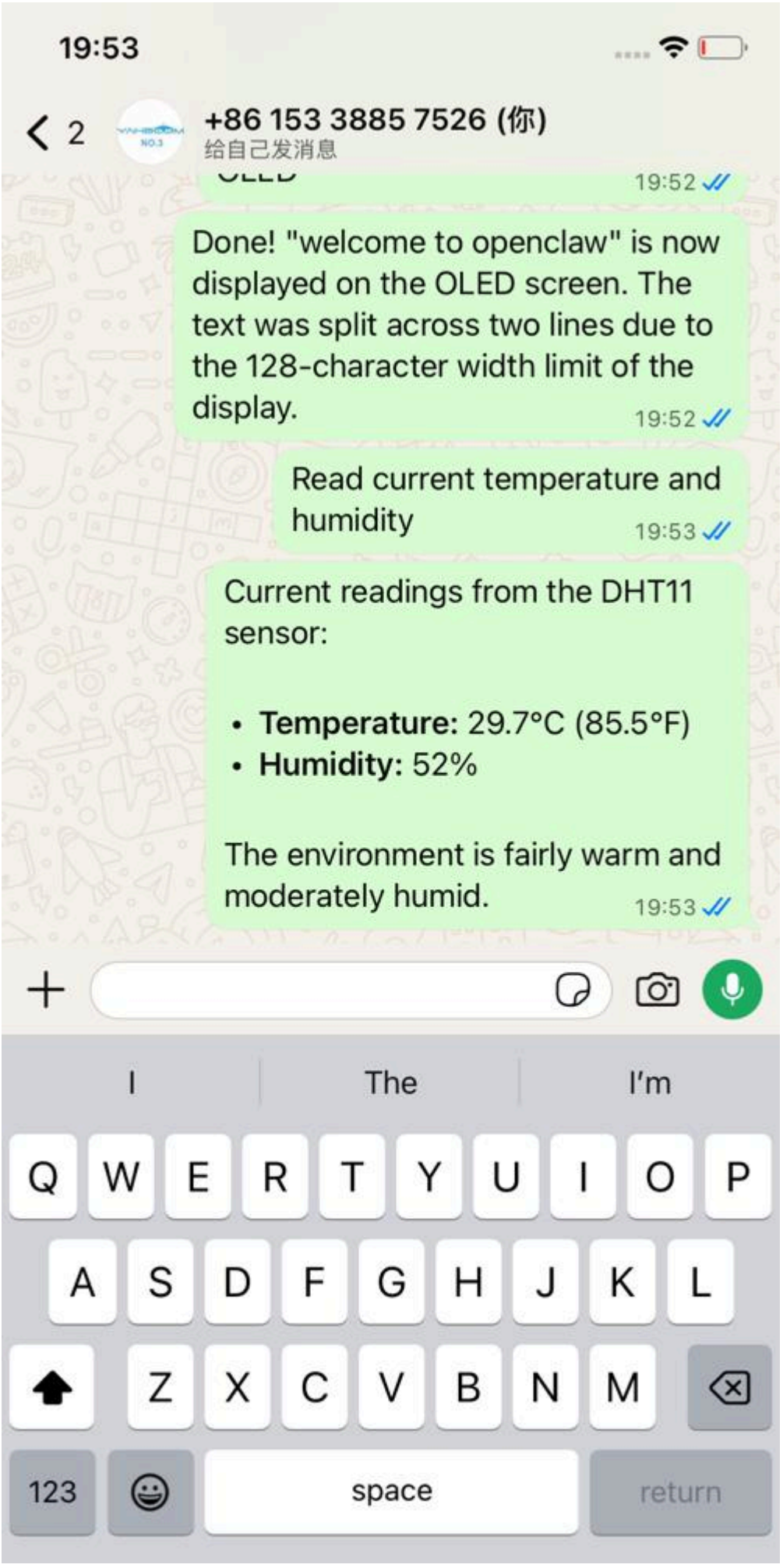


5.1 WhatsApp Scheme

For WhatsApp connection steps, please directly refer to [06-Three Dialogue Configuration Tests]. This section only includes dialogue examples suitable for the DHT tutorial.

After creation, you can directly converse, for example:

- Read the current temperature and humidity
- Judge whether it's a bit hot right now
- Alert me if the humidity is too low

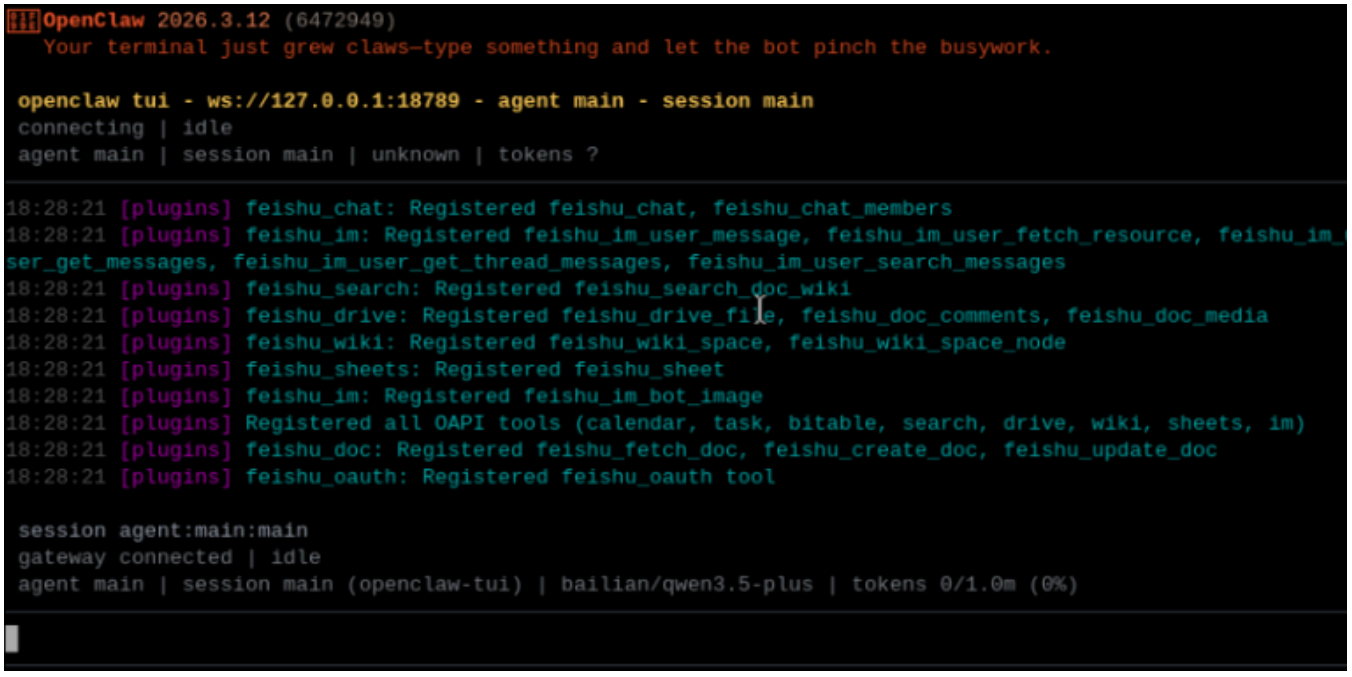

19:53
2
+86 153 3885 7526 (你)
给自己发消息
19:52
Done! "welcome to openclaw" is now displayed on the OLED screen. The text was split across two lines due to the 128-character width limit of the display.
19:52
Read current temperature and humidity
19:53
Current readings from the DHT11 sensor:
• Temperature: 29.7°C (85.5°F)
• Humidity: 52%
The environment is fairly warm and moderately humid.
19:53
I
The
I'm
Q W E R T Y U I O P
A S D F G H J K L
Z X C V B N M
123
space
return



5.2 TUI Scheme

If you have completed the TUI self-check in Part 0, simply enter the following command in the terminal to continue the conversation:

```
openclaw tui
```



5.3 Voice Code Scheme

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up process, please directly refer to [05-AI Voice Interaction Configuration].

You need to say "Hi, yahboom" to wake up for conversation.

After wake-up, you can directly speak control commands, for example:

- What is the current temperature
- Help me read the humidity
- Tell me if it's too stuffy



6. Comprehensive Cases

Note: The following are all demonstrated using the `openclaw tui` method. You can also choose WhatsApp or voice methods to complete them.

6.1 Case 1: Single Reading

Experimental Objective

Obtain a single reading of the current temperature and humidity to complete the most basic environmental sampling.

Effect Description

After OpenClaw receives the instruction, it will call the DHT driver to read the current temperature and humidity, and return the results to the user in natural language.

Example Instructions

- Read the current temperature and humidity
- What is the current indoor temperature
- Help me check the current humidity

6.2 Case 2: Comfort Judgment

Experimental Objective

Have OpenClaw make a simple judgment about the current environment after reading the data.

Effect Description

In addition to just reporting numbers, OpenClaw can also roughly judge states like "too hot", "too cold", "too dry", or "too stuffy" based on temperature and humidity, making the output sound more like a natural conversation.

Example Instructions

- Help me judge whether the current environment is comfortable
- Is the current temperature and humidity suitable for long-time office work
- Give me a brief evaluation based on the temperature and humidity

6.3 Case 3: Threshold Alert

Experimental Objective

Have OpenClaw alert the user when the temperature or humidity exceeds a set range.

Effect Description

When the temperature is too high or too low, or the humidity is too high or too low, OpenClaw can directly output alert messages, or link with RGB lights for more obvious feedback.

Example Instructions

- Alert me if the temperature exceeds 30 degrees
- Notify me if the humidity drops below 40%
- Has the alarm condition been reached

Extensible Ideas

- Link with RGB for red-yellow-blue three-color alerts
- Link with OLED to display alarm text
- Add a buzzer for stronger alerts

6.4 Case 4: OLED Linked Display

Experimental Objective

Display the just-read temperature and humidity results directly on the OLED, forming a complete data display chain.

Effect Description

After OpenClaw completes an environmental sampling, it can not only reply the results to the user but also simultaneously write the temperature and humidity to the OLED screen, forming a complete "reading + display" effect.

Example Instructions

- Read the current temperature and humidity and display on OLED
- Write the temperature and humidity to the screen
- Create a simple temperature and humidity display page

Extensible Ideas

- Periodically refresh OLED content
- Create a temperature and humidity dashboard
- Link with a weather station project to form a complete application